

Antiepileptic Drugs (AEDs): Core Pharmacology & Clinical Comparison

(1) Na⁺ Channel Blockers (Phenytoin, Carbamazepine, Lamotrigine, Topiramate, Valproate), (2) Ca²⁺ Channel Blockers (Ethosuximide, Gabapentin, Pregabalin), (3) GABA Enhancers (Valproate, Tiagabine, Gabapentin), and (4) SV2A Ligand (Levetiracetam).

Drug Name	Mechanism of Action (MoA)	Therapeutic Uses (Clinical Indications)	Adverse Effects (ADRs)	Drug Interactions & Kinetics
Phenytoin (Diphenylhydantoin)	- Blocks voltage-gated Na⁺ channels - Prolongs inactivated state of neuronal Na⁺ channels - Reduces neuronal excitability - Fosphenytoin: Soluble prodrug for IV use	- Generalized Tonic-Clonic Seizures (GTCS) - Simple Partial Seizures (SPS) - Complex Partial Seizures (CPS) • Status Epilepticus (slow IV) • Migraine	• Ataxia, Nystagmus, Diplopia • Cognitive impairment • Gingival hyperplasia ● Osteoporosis • Hirsutism ● Peripheral neuropathy • Fetal Hydantoin Syndrome (teratogenic: microcephaly, cleft palate, heart defects)	• Strong CYP450 Inducer (decreases level of co-administered drugs) • Non-linear (dose-dependent zero-order) kinetics
Carbamazepine	• Blocks voltage-gated Na⁺ channels • Inhibits high-frequency repetitive neuronal firing	- Partial Seizures (SPS, CPS) - Generalized Tonic-Clonic (GTCS) - Trigeminal Neuralgia • Acute Mania & Bipolar disorder (alternative to Lithium)	• Diplopia, Cognitive dysfunction • Blood Dyscrasias (aplastic anemia, agranulocytosis, leukopenia) • Stevens-Johnson Syndrome (SJS) • Water retention & Hyponatremia (SIADH / enhances ADH) • Teratogenic (spina bifida, craniofacial defects)	• Potent CYP450 Enzyme Inducer • Auto-induction of its own metabolism • Multiple significant drug-drug interactions (DDI)
Divalproex / Valproic Acid / Sodium Valproate	• Converts to valproate ion in body • Increases GABA levels in brain • Blocks voltage-dependent Na⁺ channels • Blocks T-type Ca²⁺ currents • Inhibits GABA-transaminase (Divalproex improves GIT tolerance)	• Drug of Choice for: - Myoclonic & Atonic Seizures • Highly effective in Absence Seizures • Alternative/Adjuvant for GTCS, SPS, CPS • Bipolar disorder & Mania (alternative to Lithium)	• Hepatotoxicity (liver failure risk) • Weight gain • Alopecia (hair loss) • Teratogenic (Spina Bifida / neural tube defects)	• CYP450 Enzyme Inhibitor (increases plasma levels of other drugs like Lamotrigine) • Significant DDIs
Ethosuximide	• Blocks T-type Calcium (Ca²⁺) channels in thalamic neurons	• ONLY effective in Absence Seizures (First-line choice)	• Weight gain • SLE-like syndrome (Systemic Lupus Erythematosus / butterfly rash) • Blood Dyscrasias (bone marrow suppression: thrombocytopenia, leukopenia, aplastic anemia) • Headache, Dizziness, Sedation	• Regular Complete Blood Count (CBC) monitoring required • Selective indication for absence seizures

Lamotrigine	- Blocks voltage-gated Na⁺ channels - Blocks High-Voltage Dependent Ca²⁺ channels	Drug of First Choice for Simple Partial Seizures - Generalized Seizures & Absence Seizures - Lennox-Gastaut Syndrome - Bipolar Disorders	Stevens-Johnson Syndrome (SJS) (severe cutaneous adverse reaction) - Serious life-threatening skin rashes	- P450 Inducers increase its metabolism - Divalproex (P450 inhibitor) inhibits its metabolism, causing high levels Lamotrigine + Divalproex combination causes SERIOUS RASH
Levetiracetam	Binds to Synaptic Vesicle Protein 2A (SV2A) to inhibit neurotransmitter release	- Adjunct treatment for: - Simple Partial Seizures (SPS) - Generalized Tonic-Clonic (GTCS) - Myoclonic Seizures	Mood alterations (irritability, agitation, depression)	- Excreted in urine mostly unchanged - NO significant CYP450 Drug-Drug Interactions (DDI)
Topiramate	- Blocks Na⁺ channels - Blocks L-type Ca²⁺ channels - Blocks Glutamate AMPA/kainate receptors - Carbonic Anhydrase Inhibitor	- Simple Partial Seizures - Generalized Seizures - Migraine prophylaxis	- Weight loss - Oligohidrosis (decreased sweating) & hyperthermia - Renal stones (kidney stones) - Paresthesia (tingling sensation) - Glaucoma, Somnolence	- Weak carbonic anhydrase inhibition - Metabolic acidosis risk when combined with other carbonic anhydrase inhibitors
Gabapentin	- Inhibits voltage-gated Ca²⁺ channels (alpha-2-delta subunit) - Increases GABA release in brain	- Post-Herpetic Neuralgia - Diabetic Neuralgia / Neuropathy - Migraine - Simple Partial Seizures (alternative) - Excellent choice in elderly patients	Very few ADRs (well-tolerated, mild sedation/dizziness)	- Few Drug-Drug Interactions (DDI) - Nonlinear pharmacokinetics (saturable absorption)
Pregabalin	- Inhibits voltage-gated Ca²⁺ channels (alpha-2-delta site) - Inhibits release of excitatory neurotransmitters	- Post-Herpetic Neuralgia - Diabetic Peripheral Neuropathy - Fibromyalgia - Simple Partial Seizures (alternative)	Well-tolerated with few ADRs (dizziness, somnolence)	90% excreted unchanged by kidneys - No significant effect on liver metabolism - Few Drug-Drug Interactions (DDI)
Tiagabine	- Depresses GABA Transporter-1 (GAT-1) - Inhibits GABA reuptake, increasing GABA levels in synapses	Adjunctive therapy for Partial Seizures	Dizziness, somnolence, nervousness, confusion	Metabolism enhanced by hepatic enzyme-inducing AEDs (phenytoin, carbamazepine)

• **Mnemonic 'PHENY' for Phenytoin ADRs:** P = Peripheral neuropathy, H = Hirsutism, E = Enlarged gums (gingival hyperplasia), N = Nystagmus/Ataxia, Y = Yields teratogenic Fetal Hydantoin Syndrome.

• **Enzyme Inducers vs. Inhibitors:** Phenytoin & Carbamazepine are CYP450 Inducers (decrease drug levels). Valproate is a CYP450 Inhibitor (increases drug levels, notably boosting Lamotrigine levels and risking severe skin reactions/SJS).

• **Absence Seizure Specificity:** Ethosuximide selectively blocks T-type Ca²⁺ channels and is ONLY effective in absence seizures (remember 'ESME' avatar: Weight gain, SLE-like rash, Blood dyscrasias). Valproate is broad-spectrum and also effective in absence seizures.

• **Safe Options with Minimal Interactions:** Levetiracetam (SV2A ligand), Gabapentin, and Pregabalin have minimal to no CYP450 drug interactions, making them ideal adjunctive agents and safer in elderly patients.